

Irrationality after One Stabilization of Universally CH_0 -Trivial Cubic Threefolds

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Abstract

Does universal CH_0 -triviality force rationality after stabilization? We construct a non-isotrivial one-parameter family of smooth cubic threefolds X_b for which X_b is universally CH_0 -trivial but $X_b \times \mathbf{P}^1$ is irrational. More generally, we prove that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold X .

The examples arise from the nonstandard A_5 -invariant cubic pencil. Six elliptic quotients of its intermediate Jacobians give an isogeny of polarization type $(1, 6, 6, 6, 6)$. Its two-primary kernel belongs to a Frobenius-conjugate pair of semisimple gluings, while its three-primary kernel is scalar. A local mixed-adjugate argument therefore puts the primitive class $\Theta^4/4!$ of the intermediate Jacobian in the integral divisor-product lattice. Voisin’s criterion gives universal CH_0 -triviality fibre by fibre.

For irrationality, we use the formal-monodromy atom of the big quantum connection. The cubic atom has primitive sixth-root eigenvalues. The projective-bundle formula gives two copies on $X \times \mathbf{P}^1$, whereas the atoms of every point, curve, or surface have no primitive sixth-root formal monodromy. Additivity and weak factorization prevent cancellation. Thus the cycle-theoretic and quantum detectors diverge on the same smooth fourfolds.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; intermediate Jacobian; quantum connection; weak factorization.

1 Introduction

A smooth projective variety is universally CH_0 -trivial if the degree map

$$\deg : CH_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension $L/\mathbf{C}$. This condition is a central consequence of stable rationality, and its failure is a standard stable-rationality obstruction. Its validity, however, need not provide a rational parametrization. We ask how far that gap can persist after one explicit stabilization.

Our examples are fourfolds of the form $Y = X \times \mathbf{P}^1$, where X is a smooth cubic threefold in the nonstandard A_5 -invariant pencil. Two independent mechanisms act on the same Y . The intermediate Jacobian of X has an algebraic primitive minimal class; this gives universal CH_0 -triviality. The big quantum connection of every smooth cubic contains a formal-monodromy atom with primitive sixth roots; weak factorization shows that this atom survives the product with $\mathbf{P}^1$.

Here “nonstandard” has a precise representation-theoretic meaning. Let W_5 be the irreducible five-dimensional complex representation of A_5 , with character $(5, 1, -1, 0, 0)$ on the classes $(1, 2A, 3A, 5A, 5B)$. The symmetric-cube character formula gives

$$\dim(\mathrm{Sym}^3 W_5^*)^{A_5} = 2.$$

Thus $\mathbf{P}((\mathrm{Sym}^3 W_5^*)^{A_5}) \cong \mathbf{P}^1$ is a pencil of cubic hypersurfaces in $\mathbf{P}(W_5)$. We denote its smooth locus by B° . It is the complement of the discriminant in the pencil and is connected. This is Hartlieb’s irreducible-character component M_{H_1} , as opposed to the component coming from the reducible permutation representation $\mathbf{1} \oplus W_4$ [5, Lemma 5.5].

Theorem 1.1 (Separation after one stabilization). *Let $\mathcal{X} \rightarrow B^\circ$ be the smooth family in the nonstandard A_5 -invariant pencil of cubic threefolds. For every $b \in B^\circ(\mathbf{C})$,*

$\mathcal{X}_b$ is universally CH_0 -trivial, $\mathcal{X}_b \times \mathbf{P}^1$ is universally CH_0 -trivial and irrational.

The family is non-isotrivial in coarse moduli.

The irrationality statement does not depend on the symmetry of the pencil.

Theorem 1.2 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

We do not claim that X is stably irrational. From the second stabilization onward, weak-factorization centers can themselves have cubic dimension, and the low-dimensional exclusion used here no longer applies.

How the two proofs meet

The cycle-theoretic half starts from six elliptic quotients of the Fano surface. Their Rosati Gram matrix is $6I - J$, so any five axes give an isogeny $E^5 \rightarrow J(X)$ of polarization type $(1, 6, 6, 6, 6)$. The two-primary principal kernels form $\mathbf{P}^1(\mathbf{F}_4)$. Three retain the full simplex symmetry; the other two are exchanged by Frobenius and have stabilizer A_5 . The cycle argument needs only this unordered exotic pair. At two its graph slope has minimal polynomial $t^2 + t + 1$; at three every A_5 -stable graph is scalar. Both local forms admit primitive mixed-adjugate formulas, which produce $\Theta^4/4!$ as an integral product of divisor classes.

The quantum half uses a different invariant. We count primitive sixth-root eigenvalues of formal monodromy on the atoms of the maximal big-quantum F -bundle. Cai’s cubic calculation gives a nonzero count. The projective-bundle formula doubles it on $X \times \mathbf{P}^1$. If this fourfold were rational, weak factorization would express that count using centers of dimension at most two. Surface classification and the low-dimensional atom formula show that every such center has count zero.

Relation with earlier rationality results

Clemens and Griffiths proved that every smooth complex cubic threefold is irrational by means of its intermediate Jacobian [3]. That obstruction does not by itself control a product with projective space, and stable rationality of smooth cubic threefolds remains open. Cai recently recovered irrationality from the primitive-sixth-root formal monodromy of the cubic quantum connection [2]. Katzarkov–Kontsevich–Pantev–Yu supply the atom formalism, blowup and projective-bundle formulas, and the resulting nonrationality criterion [6]. Theorem 1.2 combines these inputs at the next dimension: after one projective stabilization, the cubic atom still cannot be contributed by any weak-factorization center.

On the cycle side, Voisin reduces universal CH_0 -triviality of a cubic threefold to algebraicity of the primitive minimal class of its intermediate Jacobian [9]. Beckmann–de Gaay Fortman place such minimal classes in a general integral-Fourier framework [1]. Here the six elliptic quotients supplied by Roulleau and the special period locus isolated by Hartlieb make the class explicit: the new point is its membership in the ordinary integral divisor-product lattice for every smooth member of the nonstandard A_5 -pencil.

Proof map

Section 2 constructs the rank-ten symplectic envelope and identifies the exotic cubic gluing. Section 3 proves the local divisor-product theorem and applies it to the primitive minimal class. Section 4 proves Theorem 1.2 from formal monodromy and weak factorization. The final section combines the two detectors and records the exact scope boundary.

2 The integral envelope and the exotic cubic sheet

Let Ω be the six conjugate D_5 -subgroups of A_5 , equivalently the six Sylow-5 subgroups with their normalizers. Put

$$A(\Omega) = \ker(\mathbf{Z}^\Omega \xrightarrow{\Sigma} \mathbf{Z}), \quad A(\Omega)^\vee = \operatorname{Hom}(A(\Omega), \mathbf{Z}).$$

The perfect evaluation pairing defines a unimodular alternating form on

$$\mathcal{H}_{\text{ref}}(\Omega) = A(\Omega)^\vee e \oplus A(\Omega)f, \quad \langle xe, yf \rangle = y(x).$$

This is the hyperbolic symplectic envelope of the oriented six-set.

Proposition 2.1 (Universal envelope). *The construction $\Omega \mapsto \mathcal{H}_{\text{ref}}(\Omega)$ is functorial for permutations of Ω . Rationally it is the unique rank-ten symplectic A_5 -module whose two Lagrangians are copies of the standard five-dimensional constituent and whose pairing restricts to the root-weight evaluation pairing.*

Proof. The permutation lattice and its augmentation kernel are functorial. The evaluation pairing between $A(\Omega)$ and its dual is perfect, so the displayed alternating form is unimodular. Over $\mathbf{Q}$, the augmentation module is absolutely irreducible and its invariant symmetric form is unique up to scale. Schur's lemma therefore writes every symplectic module with two such constituents as $A(\Omega)\mathbf{Q} \otimes U$, where U is a symplectic plane. Requiring the two integral Lagrangians to pair perfectly fixes the root-weight lattice above. $\square$

The finite golden orientation enters at the prime two. Let

$$H_2 = \operatorname{Aug}(\mathbf{F}_2^\Omega) / \langle \mathbf{1} \rangle.$$

Then H_2 is simple and $\operatorname{End}_{A_5}(H_2) = \mathbf{F}_4$. A normalized conference operator acts integrally on $\mathbf{Z}^\Omega$. After labeling Ω , one switching representative is

$$B = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & -1 & -1 \\ 1 & 1 & 0 & -1 & 1 & -1 \\ 1 & 1 & -1 & 0 & -1 & 1 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & -1 & 1 & 1 & 0 \end{pmatrix}, \quad B^2 = 5I.$$

Passing to the canonical saturation

$$D_6^\vee = \mathbf{Z}^\Omega + \mathbf{Z}\frac{\mathbf{1}}{2}, \quad \varphi = \frac{I+B}{2},$$

gives $\varphi^2 - \varphi - 1 = 0$. Its reduction on the commutator heart is one of the two primitive scalars $\omega, \omega^2 \in \mathbf{F}_4$, and replacing B by $-B$ applies Frobenius.

Proposition 2.2 (Golden selection of the exotic pair). *The two orientations of the conference class determine the two primitive scalars in $\text{End}_{A_5}(H_2) = \mathbf{F}_4$. Under orientation reversal,*

$$B \longmapsto -B, \quad \omega \longmapsto \omega^2.$$

Thus the golden-orientation torsor is canonically the Frobenius torsor of the exotic two-primary principal gluings.

Proof. Every column of $(I + B)/2$ has the same half-integral class modulo $\mathbf{Z}^\Omega$, so $D_6^\vee$ is the least stable over-lattice. The commutator submodule of $D_6^\vee/2D_6^\vee$ is the four-dimensional six-point heart and its quotient is a trivial $\mathbf{F}_4$ -line. The extension has no invariant $\mathbf{F}_4$ -line, so the restriction of $\bar{\varphi}$ to the heart is primitive. Finally $(I - B)/2 = 1 - \varphi$, and in characteristic two $1 - \omega = \omega^2$. $\square$

We now recall the geometric source of the same lattice. Let S be the Fano surface of a smooth member of the A_5 -pencil and $(J, \Theta) = \text{Alb}(S)$ its intermediate Jacobian. Each of the six D_5 -subgroups gives an elliptic quotient $q_H : J \rightarrow E_H$ and its Rosati dual $i_H : E_H \rightarrow J$.

Proposition 2.3 (Six-axis polarization). *After coherent identification of the elliptic factors, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

Any five axes therefore give an isogeny $E^5 \rightarrow J$ whose pulled-back polarization has Smith type $(1, 6, 6, 6, 6)$.

Proof. We recall the numerical passage from Roulleau's genus-two configuration to the six elliptic fibres. Let $D_g \subset S$ be the curve indexed by an involution $g \in A_5$, put $D_{\text{tot}} = \sum_g D_g$, and, for a dihedral subgroup H of order ten, put

$$F_H = \sum_{g \in H, g^2=1} D_g.$$

Roulleau proves that F_H is a fibre of $S \rightarrow E_H$ [8, Theorem 11(D)], so $F_H^2 = 0$. His intersection table has $D_g^2 = -4$, and distinct curves meet with multiplicity two or one when the product of their labels has order three or five, respectively. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur 15, 30, 60 times. Hence

$$D_{\text{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore $\sum_H F_H = 2D_{\text{tot}}$. Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\text{tot}})^2 = 720, \quad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

After coherent identification of the generic non-CM elliptic quotients, two-transitivity makes the Rosati matrix have diagonal d and off-diagonal m . Put

$$D_H = q_H^*[0] \in \text{NS}(J), \quad N_H = i_H q_H \in \text{End}(J).$$

Then $D_H|_S = F_H$, and N_H is the Rosati endomorphism attached to D_H . Coherent identification gives $q_H i_H = [d]$ and $q_H i_{H'} = [m]$ for $H \neq H'$. The invariant sum of the six axis

maps vanishes because the five-dimensional constituent has no trivial summand; composing with q_H gives $d + 5m = 0$. The minimal-class identity $[S] = \Theta^3/3!$, together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \mathrm{tr}(N_\alpha)\mathrm{tr}(N_\beta) - \mathrm{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [4, Theorem 4.1], gives

$$\mathrm{tr}(N_H) = d, \quad \mathrm{tr}(N_H N_{H'}) = m^2, \quad F_H F_{H'} = d^2 - m^2.$$

Thus $d^2 - m^2 = 24$. The positive solution is $(d, m) = (5, -1)$, hence $G_6 = 6I - J$. A five-by-five minor has Smith form $(1, 6, 6, 6, 6)$. $\square$

Proposition 2.4 (Principal gluing packet). *The A_5 -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

The three $\mathbf{F}_2$ -rational kernels are S_6 -stable. Each of the other two has stabilizer exactly A_5 , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard A_5 -cubic component uses the latter pair. Every A_5 -stable three-primary kernel is a scalar graph; the $\Gamma_0(3)$ multiplicity monodromy selects one of them.

Proof. At two the coefficient discriminant is the simple four-dimensional $\mathbf{F}_2 A_5$ -module H_2 , whose commutant is $\mathbf{F}_4$. The full discriminant is $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2$. An A_5 -stable simple half is therefore a one-dimensional $\mathbf{F}_4$ -subspace of $\mathbf{F}_4^2$, and every such half is maximal isotropic for the tensor product of the coefficient form with the alternating multiplicity form. This gives $\mathbf{P}^1(\mathbf{F}_4)$.

Here is the polarization check explicitly. Identify the coefficient heart with $\mathbf{F}_4^2$ and put

$$b(x, y) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar $a \in \mathbf{F}_4$ is self-adjoint for b , since $b(ax, y) = b(x, ay)$. Thus the vertical line and all four graphs Γ_a are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over $\mathbf{F}_2$. Hence the three lines defined over $\mathbf{F}_2$ retain S_6 . The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes Γ_ω commutes with ω , hence is $\mathbf{F}_4$ -linear on the two-dimensional $\mathbf{F}_4$ -space H_2 . If its determinant is $d \in \mathbf{F}_4^\times$, preservation of b says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \quad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so $d = 1$. It therefore lies in $\mathrm{SL}_2(4)$, of order 60. The given A_5 already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges ω and ω^2 . Were the generic cubic kernel $\mathbf{F}_2$ -rational, its principal polarization would inherit the faithful S_6 -action. Strong Torelli identifies $\mathrm{Aut}(J, \Theta)$ with $\mathrm{Aut}(X) \times \langle -1 \rangle$. Projection of this S_6 to $\mathrm{Aut}(X)$ remains faithful: its kernel would be a normal subgroup of S_6 of order at most

two, and no such nontrivial subgroup exists. This contradicts the generic automorphism group A_5 on the nonstandard component [5, Section 5]. Thus the generic kernel is exotic. The three rational kernels and the two exotic kernels are distinguished by their stabilizers under the fibrewise A_5 -action; they are therefore distinct monodromy-stable subsets of the five-element gluing packet. The geometric kernel is a finite sub-local-system, so generic exoticity and the connectedness of the smooth locus force it to remain in the exotic pair on every fibre.

At three the coefficient heart H_3 is simple and $\text{End}_{A_5}(H_3) = \mathbf{F}_3$. Hence its A_5 -stable halves are the four graphs parametrized by $\mathbf{P}^1(\mathbf{F}_3)$, all scalar in a suitable graph chart. $\square$

Proposition 2.2 labels the two exotic members by the finite golden orientation.

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial $t^2 + t + 1$. The golden orientation supplies the finer marked identification with one member of the pair, but Theorem 3.6 uses only the conclusion that the geometric kernel is not one of the three $\mathbf{F}_2$ -rational gluings. At three it uses only scalarity of the possible A_5 -stable graphs, not the finer monodromy selection.

Hartlieb proves that its intermediate-Jacobian image is a one-dimensional special subvariety [5, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

3 The primitive minimal class

Let (A, Θ) be a principally polarized abelian variety of dimension g . Write

$$\gamma_A = \frac{\Theta^{g-1}}{(g-1)!} \in H^{2g-2}(A, \mathbf{Z}).$$

The issue is not the rational algebraicity of γ_A , but its membership in the integral lattice generated by ordinary products of divisor classes.

We use one standard graph chart explicitly. Let E be an elliptic curve, give E^g the coefficient polarization pI_g , and choose symplectic coordinates on $E[p]$. A symmetric matrix $T \in \text{Sym}_g(\mathbf{F}_p)$ defines the maximal isotropic subgroup

$$K_T = \{(x, Tx) : x \in \mathbf{F}_p^g\} \subset E[p]^g.$$

The quotient E^g/K_T carries the descended principal polarization.

Lemma 3.1 (The graph descent congruence). *Choose a symmetric integral lift $\tilde{T}$ of T . For $D \in \text{Sym}_g(\mathbf{Z}_p)$, the coefficient divisor pD on the source descends through the graph quotient whenever*

$$D\tilde{T} - \tilde{T}D \equiv 0 \pmod{p}.$$

In particular, on a summand on which the graph slope is scalar modulo p , every block-supported coefficient in $p\text{Sym}_d(\mathbf{Z}_p)$ descends.

Proof. Write the source homology as $L_e \oplus L_f$, with its standard alternating coordinates. The quotient homology is generated over the source by

$$p^{-1}(x, \tilde{T}x), \quad x \in L.$$

The alternating form of the coefficient divisor pD pairs an integral generator with each displayed graph generator integrally. On two graph generators its value is

$$\frac{1}{p}x^\dagger(D\tilde{T} - \tilde{T}D)y.$$

The stated congruence is therefore exactly the remaining integrality condition. If the slope is scalar on a block, the commutator vanishes there modulo p . $\square$

We first isolate the normalization that will be used throughout. If $C_1, \dots, C_d$ are symmetric endomorphisms of a free module, write

$$\text{Mdet}(C_1, \dots, C_d) = [t_1 \cdots t_d] \det\left(\sum t_i C_i\right)$$

and define $\text{Madj}(C_1, \dots, C_{d-1})$ in the same way from the adjugate. Thus these are coefficient operators, not derivatives divided by a factorial.

Lemma 3.2 (Primitive mixed coefficients). *On $\text{Sym}_d(\mathbf{Z})$, the mixed determinant has image $\mathbf{Z}$, and the entries of the mixed adjugates span $\text{Sym}_d(\mathbf{Z})$. More precisely,*

$$\text{Mdet}(E_{11}, \dots, E_{dd}) = 1,$$

while the coefficient of $\prod_{r \neq i} t_r$ in $\text{adj}(\sum_{r \neq i} t_r E_{rr})$ is E_{ii} , and inserting $E_{ij} + E_{ji}$ together with the E_{rr} , $r \neq i, j$, gives $-E_{ij} - E_{ji}$.

Proof. The first two assertions are the diagonal monomials in the Leibniz formulas for the determinant and its cofactors. For the last assertion the only surviving permutations use the transposition (ij) , whose sign is minus. Every displayed coefficient is therefore ± 1 . In particular neither a factorial nor the usual factor two for an off-diagonal quadratic term occurs. $\square$

The determinant is also the polarization dictionary. If a polarization on an elliptic power has coefficient matrix G , normalized Riemann–Roch identifies its top intersection polynomial with $\det G$ [4, Theorem 4.1 and Corollary 4.2]. Polarizing that identity shows that a product of $g - 1$ coefficient divisors is represented by the corresponding mixed adjugate. In particular, the pullback of the primitive minimal class is represented by $\text{adj}(G)$. This is why the coefficient-one statements in Lemma 3.2 are the exact integral normalization needed below.

Theorem 3.3 (Semisimple graph slopes). *Let p be a prime and let $T \in \text{Sym}_g(\mathbf{F}_p)$. Form the principally polarized graph quotient above. If the minimal polynomial of T is squarefree, then*

$$\gamma_A \in \text{im}(\text{Sym}^{g-1} \text{NS}(A) \longrightarrow H^{2g-2}(A, \mathbf{Z})).$$

Proof. Work over $\mathbf{Z}_p$, and pass to a finite unramified extension over which the squarefree minimal polynomial splits. Self-adjoint eigenspaces for distinct roots are orthogonal and nondegenerate. They lift, without dividing by two, to an orthogonal decomposition with blocks pB_λ , where each B_λ is symmetric and unimodular.

On a scalar slope block, Lemma 3.1 shows that the integral divisor lattice contains every block-supported coefficient in $p\text{Sym}_{d_\lambda}(O)$. Fill every non-target block by its primitive mixed determinant from Lemma 3.2. On the target block, its degree-one-lower mixed adjugates give every diagonal matrix unit and every symmetric off-diagonal matrix unit with coefficient ± 1 . Because the blocks have disjoint support, taking one coefficient from each block simply multiplies these primitive coefficients; no multinomial factor appears.

Summing over target blocks constructs the cofactor of $\perp_\lambda pB_\lambda$, which is the pullback of γ_A . No factorial or off-diagonal factor two is introduced. Let $P \subset H^{2g-2}(A, \mathbf{Z}_p)$ be the image of the local divisor-product map. Formation of P commutes with the finite unramified extension $O/\mathbf{Z}_p$. Since O is faithfully flat, the map

$$H^{2g-2}(A, \mathbf{Z}_p)/P \longrightarrow (H^{2g-2}(A, \mathbf{Z}_p)/P) \otimes_{\mathbf{Z}_p} O$$

is injective; membership therefore descends from O . Away from p , the quotient isogeny identifies the integral divisor and cohomology lattices. The cokernel of the global divisor-product map is finitely generated, so a class lies in its image exactly when it does so after tensoring with every $\mathbf{Z}_q$. Thus local membership globalizes. $\square$

Corollary 3.4 (Unimodular coefficient form). *Suppose that, over $\mathbf{Z}_p$, the coefficient lattice is an orthogonal sum $U_0 \perp pB$, where U_0 and B are unimodular. Assume that the principal kernel is trivial on U_0 and is a graph on the defect block whose slope is self-adjoint for B and has squarefree minimal polynomial modulo p . Then the pullback of the primitive minimal class belongs to the local lattice generated by products of $g - 1$ descended divisors.*

Proof. After unramified splitting, the self-adjoint eigenspaces are mutually B -orthogonal and nondegenerate. Unimodular orthogonal complements lift them integrally without division by two. On each scalar block the invariant graph-descent condition

$$D\tilde{T} - \tilde{T}^t D \equiv 0 \pmod{p}$$

contains every block-supported symmetric coefficient pD . Identifying symmetric coefficient forms with B -self-adjoint endomorphisms, the same mixed-determinant and mixed-adjugate argument constructs the cofactor of $U_0 \perp pB$. The faithful-flat and local-global arguments in Theorem 3.3 apply unchanged. $\square$

The scalar version of the same mixed-adjugate construction applies to the three-primary block of the six-axis polarization. We can therefore use the two local forms isolated in Section 2.

Lemma 3.5 (Local chart for the six-axis quotient). *Let $G = 6I_5 - J_5$. At each prime $p \mid 6$, the coefficient lattice has an orthogonal Jordan decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where U_0 has rank one and is unimodular and U_1 has rank four and is unimodular. The principal quotient is trivial on U_0 . On U_1/pU_1 , its graph slope has minimal polynomial $t^2 + t + 1$ at $p = 2$, while at $p = 3$ every A_5 -stable graph slope is scalar on the simple coefficient constituent.

Proof. The Smith form of G is $(1, 6, 6, 6, 6)$, but here the orthogonal chart can also be written down. Let $e_1, \dots, e_5$ be the standard basis. Since $\langle e_1, e_1 \rangle = 5$ and $\langle e_1, e_j \rangle = -1$, the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of e_1 over $\mathbf{Z}_p$. In this basis its Gram matrix is

$$\frac{6}{5}(5I_4 - J_4).$$

For $p = 2, 3$, both 5 and $\det(5I_4 - J_4) = 5^3$ are units. We have therefore exhibited a unimodular line orthogonal to p times a unimodular rank-four block. The unit block has no discriminant and hence no gluing choice.

At two the discriminant heart is the six-point module H_2 , and Proposition 2.2 identifies the selected endomorphism with $\omega \in \mathbf{F}_4$; its polynomial is $t^2 + t + 1$. At three the coefficient heart is simple with scalar commutant, so every A_5 -stable half is a scalar graph. The monodromy-invariant line identifies which scalar graph occurs in the family but is not needed for primitivity. $\square$

Theorem 3.6 (Minimal class for the exotic cubic family). *For every smooth member X_b of the nonstandard A_5 -cubic pencil, with intermediate Jacobian (J_b, Θ_b) ,*

$$\frac{\Theta_b^4}{4!} \in \text{im}(\text{Sym}^4 \text{NS}(J_b) \longrightarrow H^8(J_b, \mathbf{Z})).$$

Proof. The five-axis source has polarization type $(1, 6, 6, 6, 6)$, so only the primes two and three can contribute an integral defect. At two the exotic graph slope acts on the simple heart through $\omega \in \mathbf{F}_4$; its minimal polynomial $t^2 + t + 1$ is squarefree. Theorem 3.3 via Corollary 3.4 therefore gives the primitive cofactor on the rank-four defect block. The rank-one unit block has trivial kernel, so its primitive coefficient divisor descends. Filling the unit block by its determinant when the defect block is the adjugate target, and filling the defect block by its determinant when the unit block is the target, constructs the cofactor of the full five-dimensional form. At three every A_5 -stable gluing is scalar on the simple coefficient block, and the same blockwise construction applies. Away from two and three the source and quotient lattices are unimodular. Local membership at every prime is equivalent to integral membership in the divisor-product lattice.

Lemma 3.5 identifies the actual geometric lattice, not merely an abstract gluing with the same Smith form. The graph kernel is a finite sub-local-system on each connected smooth component of the pencil, so the same local presentation holds on every fibre. At a fibre with additional Néron–Severi classes the divisor-product image can only increase. $\square$

Voisin identifies this primitive class as the exact obstruction relevant to cubic threefolds.

Corollary 3.7 (Universal CH_0 -triviality). *Every smooth member of the nonstandard A_5 -cubic pencil is universally CH_0 -trivial.*

Proof. By Theorem 3.6, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin’s Corollary 4.4 [9] says that, for a smooth complex cubic threefold, this is equivalent to universal CH_0 -triviality. $\square$

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.7.

4 The primitive-sixth-root atom

We use the geometric atomic F -bundle invariant of Katzarkov–Kontsevich–Pantev–Yu [6]. Only three features enter: its values lie in the free abelian group on atom-equivalence classes, it is additive under blowups, and a rank- r projective bundle contributes r copies of the base.

Concretely, the reduced unramified spectral cover of the Euler operator decomposes the maximal big-quantum F -bundle into connected local atoms [6, Theorem 4.1 and Definition 5.10]. If $\mathcal{A}$ denotes the set of geometric atom classes, its chemical formula has the form

$$\text{CFF}(Z) = \sum_{\alpha \in \mathcal{A}} m_\alpha(Z) [\alpha] \in \mathbf{Z}^{(\mathcal{A})}, \quad m_\alpha(Z) \geq 0.$$

The free group, rather than a quotient of a signed block ledger, is the target used below.

Definition 4.1. For a geometric atom α , restrict its connection to the original formal punctured u -disc. Let $M_f(\alpha)$ be its intrinsic formal-monodromy operator. Equivalently, after a minimal Levelt–Turruttin ramification, combine the regular-singular monodromies with the deck permutation of the exponential factors and descend to the original u -disc. Let $\text{FM}(\alpha)$ be the eigenvalue multiset of this descended operator, and define

$$\nu_6(\alpha) = \text{mult}_{e^{\pi i/3}} \text{FM}(\alpha) + \text{mult}_{e^{-\pi i/3}} \text{FM}(\alpha).$$

Extend ν_6 additively to the free atom group.

This definition uses eigenvalues in $\mathbf{C}^\times$, not chosen logarithms. It is independent of a further ramified refinement: the refined deck cycle and the refined regular monodromy compose to the same descended operator on the original differential module. Integral gauge powers shift residue exponents by integers and leave the operator unchanged up to conjugacy; see [7, Section 3.2] and [2, Remark 5] for the universal formal solution ring and formal differential-module classification.

Lemma 4.2 (Formal isomonodromy). *Along a connected component of the reduced unramified spectral cover, the formal-monodromy eigenvalue multiset of an atom is constant.*

Proof. We use flatness in all bulk directions, rather than only the Witt-leaf statement of [6, Remark 3.14]. Work locally on the analytic base and choose coordinates t_i . The big quantum connection has commuting operators

$$\nabla_{t_i} = \partial_{t_i} + u^{-1}P_i, \quad \nabla_u = \partial_u + L.$$

Solve

$$\partial_{t_i} M = -u^{-1}P_i M, \quad M|_{t=t_0} = I.$$

The compatibility equations are precisely the flatness equations for the t_i -directions, so the solution exists in the enlarged t -adic ring used by Cai, locally of the form $\text{GL}(H \otimes \mathcal{K}((u))[[t-t_0]])$, where $\mathcal{K}$ contains the Novikov coefficients. This is not an ordinary Laurent gauge over the analytic base: its $(t-t_0)$ -coefficients may have negative u -orders unbounded below. Each coefficient nevertheless involves only integral powers of u . Consequently

$$M^{-1} \nabla_{t_i} M = \partial_{t_i}.$$

Commutation with $M^{-1} \nabla_u M = \partial_u + \tilde{L}$ now gives $\partial_{t_i} \tilde{L} = 0$ for every i . This is the bulk-gauge argument used by Cai immediately before [2, Proposition 6].

After this scalar extension, the resulting gauge identifies the formal solution spaces without changing their fractional u -exponents: coefficientwise formal monodromy fixes every integral power of u . Equivalently, it preserves the eigenvalue multiset of descended formal monodromy. On the reduced unramified spectral cover the unique local spectral projector of [6, Theorem 4.1] tracks one chosen atomic factor; on overlaps the resulting identifications differ only by conjugacy. Hence its formal-monodromy characteristic polynomial is locally constant on each component and therefore constant because the component is connected. Combining the factors with their deck action gives the same statement on the original u -disc, independently of the chosen Levelt–Turruttin refinement. $\square$

Corollary 4.3 (The sixth-root multiplicity is atomic). *The function ν_6 is well defined on geometric atom-equivalence classes and extends uniquely to a homomorphism*

$$\nu_6 : \mathbf{Z}^{(\mathcal{A})} \longrightarrow \mathbf{Z}.$$

Proof. An isomorphism of geometric F -bundles in [6, Definition 5.21] is an isomorphism over the same formal u -disc, so it conjugates formal monodromy without changing its eigenvalues. Lemma 4.2 handles movement between points of one connected unramified spectral component. The blowup and projective-bundle identifications used to pass from abstract atoms to geometric atoms are likewise F -bundle isomorphisms over that same loop coordinate [6, Proposition 5.22]; no ramified substitution $u \mapsto u^m$ is among the equivalences. Definition 3.1 of [6] fixes the disc and its coordinate; base change acts on the base factor and leaves u fixed. Thus ν_6 is constant on each class, and additivity on the free group is unique. $\square$

For a smooth projective variety Z , we abbreviate $\nu_6(Z) = \nu_6(\text{CFF}(Z))$.

Proposition 4.4 (Cubic and low-dimensional atoms). *Let X be a smooth cubic threefold.*

- (i) *The cubic zero atom α_X satisfies $\nu_6(\alpha_X) \geq 2$.*
- (ii) *Every atom represented by a point, a curve, or a smooth projective surface has $\nu_6 = 0$.*

Proof. Cai's calculation preceding Proposition 6 gives the indicial polynomial

$$\rho^2 + \rho + \frac{5}{36},$$

whose roots are $-1/6$ and $-5/6$, in the rank-two zero-eigenvalue block of the even big cubic quantum connection [2]. Its bulk-variable gauges use integral powers of the loop variable. Parity preservation places this rank-two block inside the unsplit zero atom of the full maximal big-quantum F -bundle in [6, Example 6.21]; passage to that bundle is scalar extension with the same loop coordinate. Hence the two monodromy eigenvalues $e^{-\pi i/3}$ and $e^{\pi i/3}$ occur in $\text{FM}(\alpha_X)$. We make no assertion about additional eigenvalues from the odd summand, and obtain the stated lower bound.

A point has exponent zero, and $\mathbf{P}^1$ reduces to two point atoms by the projective-bundle formula. For a curve of genus $g > 0$, the explicit curve connection has formal monodromy eigenvalues only in $\{1, -1\}$: this follows directly from Cai's Proposition 7 on the even block and the classical odd block, and is also the curve-atom comparison used in [6, Example 6.21]. Thus no curve atom has a primitive sixth-root eigenvalue. For a minimal surface with nef canonical class, at an even base point with vanishing H^0 -coordinate the parity-corrected connection has nilpotent residue by [6, Claim 6.15]; undoing the integral grading and half-parity correction permits only monodromy eigenvalues 1 and -1 . A minimal surface of Kodaira dimension $-\infty$ is $\mathbf{P}^2$ or ruled. Projective space reduces to a point atom, and a ruled surface reduces by the projective-bundle formula to curve atoms. Finally, every nonminimal surface is obtained from a minimal one by point blowups, whose blowup formula adds only point atoms [6, Theorem 4.5]. The classification of smooth projective surfaces exhausts the possibilities and proves (ii). $\square$

Proof of Theorem 1.2. Let $Y = X \times \mathbf{P}^1$. The projective-bundle formula gives

$$\text{CFF}(Y) = 2 \text{CFF}(X) \quad [6, \text{Theorem 4.11 and Proposition 5.22(3)}],$$

so Proposition 4.4 gives $\nu_6(Y) > 0$.

Suppose that Y were rational. Weak factorization of a birational map $Y \dashrightarrow \mathbf{P}^4$, together with the blowup chemical formula [6, Proposition 5.22(2)], gives in the free abelian atom group

$$\text{CFF}(Y) + \sum_i (r_i - 1) \text{CFF}(Z_i) = \text{CFF}(\mathbf{P}^4) + \sum_j (s_j - 1) \text{CFF}(W_j),$$

where r_i, s_j are the corresponding codimensions. Divisorial blowups are trivial, so every effective center has codimension at least two and hence dimension at most two. Apply ν_6 . Proposition 4.4(ii) kills every center term, and the projective-bundle formula over a point gives $\nu_6(\mathbf{P}^4) = 0$. The left-hand side retains $\nu_6(Y) > 0$, a contradiction. $\square$

The proof uses the free atom group, not a signed informal block ledger. In particular the two copies supplied by $\mathbf{P}^1$ are positive multiplicities and cannot cancel one another.

5 The separation theorem

Proof of Theorem 1.1. Corollary 3.7 gives universal CH_0 -triviality of X_b . The projective bundle formula for Chow groups gives the same property for $X_b \times \mathbf{P}^1$. Theorem 1.2 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the A_5 -component. $\square$

The theorem separates two tests on one variety. The algebraic class $\Theta^4/4!$ removes the decomposition-of-diagonal obstruction. The primitive-sixth-root atom survives one stabilization and obstructs a rational factorization. The first mechanism is special to the exotic A_5 -family; the second holds for every cubic threefold.

The result stops after one stabilization. For $X \times \mathbf{P}^m$ with $m \geq 2$, weak-factorization centers may have dimension at least three and can carry cubic-type atoms. The low-dimensional exclusion in Proposition 4.4 no longer proves noncancellation. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

Thus the same smooth fourfolds pass the universal CH_0 -triviality test and fail the quantum-atomic rationality test. Vanishing of the decomposition-of-the-diagonal obstruction does not predict rationality even after one explicit stabilization.

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